MATH 152 **QUIZ 11A** NAME Solutions

Mrs. Bonny Tighe 25 points

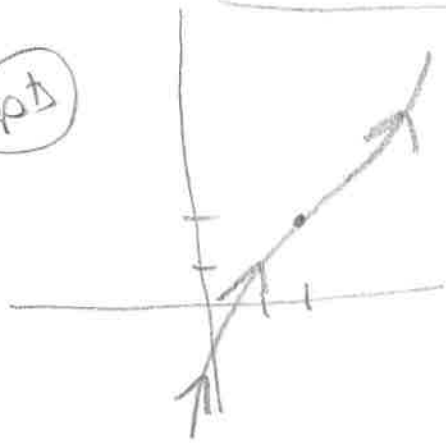
10.1, 10.2,10.3 TA_____Wed 12/4/19

1. Sketch the plane curve defined by the given parametric equations. Indicate with an arrow the direction in which the curve is traced as the parameter increases. Eliminate the parameter to find a Cartesian equation of the curve.

a) $x = t + 2, \quad y = t^3 + 2$

$t = (x-2)$ $\boxed{y = (x-2)^3 + 2}$

2pts

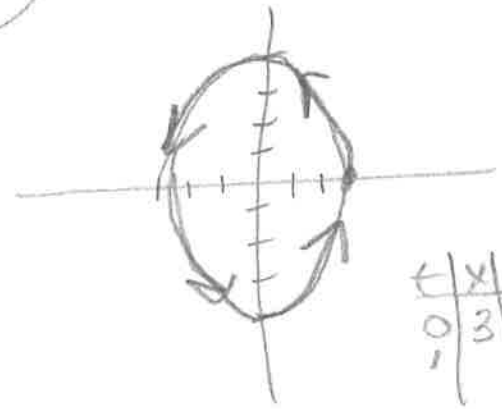


t	x	y
0	2	2
1	3	3

b) $x = 3\cos\theta, \quad y = 4\sin\theta$

$\boxed{\frac{x^2}{9} + \frac{y^2}{16} = 1}$

2pts



t	x	y
0	3	0

2. Find a parametric equation for the ellipse $\frac{y^2}{64} + \frac{x^2}{25} = 1$ if it travels clockwise.

1pt

$x = 5\sin\theta$

$y = 8\cos\theta$

3. Find the equation of the tangent line to the curve $x = 2 + \ln t, \quad y = t^2 - 2$ at (2, -1)

$x = 2 \quad y = -1$

so $t = 1$

$dy/dx = \frac{dy/dt}{dx/dt} = \frac{2t}{\frac{1}{t}} = 2t^2$

$m = 2$

1pt

$\boxed{y + 1 = 2(x-2)}$

$x=0$

4. Find the area enclosed by the y- axis and the curve $x = 1 - t^2$, $y = t - t^2$

$0 = (1+t)(1-t)$

$t = \pm 1$

2pts

$$\int_{-1}^{1} x\,dy = \int_{-1}^{1} (1-t^2)(1-2t)\,dt = \int_{-1}^{1} (1 - 2t - t^2 + 2t^3)\,dt$$

$$t - t^2 - \tfrac{1}{3}t^3 + \tfrac{1}{2}t^4 \Big]_{-1}^{1} = \left(1 - 1 - \tfrac{1}{3} + \tfrac{1}{2}\right) - \left(-1 - 1 + \tfrac{1}{3} + \tfrac{1}{2}\right)$$

$$-\tfrac{1}{3} + \tfrac{1}{2} + 2 - \tfrac{1}{3} - \tfrac{1}{2}$$

$$2 - \tfrac{2}{3} = 4/3$$

5. Compute the surface area obtained by rotating the curve

3pts

$x = 3t - t^3$, $y = 3t^2$, $0 \le t \le 1$ about the x-axis. $S = 2\pi \int_{\alpha}^{\beta} radius(arc\ length)$

$$2\pi \int y \sqrt{(dx/dt)^2 + (dy/dt)^2}\,dt$$

$$2\pi \int_0^1 3t^2 \sqrt{(3-3t^2)^2 + (6t)^2}\,dt$$

$$2\pi \int_0^1 3t^2 \sqrt{9 - 18t^2 + 9t^4 + 36t^2}\,dt$$

$$2\pi \int_0^1 3t^2 \sqrt{9 + 18t^2 + 36t^4}\,dt = 2\pi \int_0^1 3t^2 \sqrt{(3+3t^2)^2}\,dt$$

$$2\pi \int_0^1 3t^2 (3 + 3t^2)\,dt = 2\pi \int_0^1 (9t^2 + 9t^4)\,dt$$

$$2\pi \left(3t^3 + \tfrac{9}{5}t^5\right)\Big|_0^1 = 2\pi \left[3 + \tfrac{9}{5}\right] = 2\pi \left(\tfrac{24}{5}\right)$$

$x = r\cos\theta \qquad y = r\sin\theta$ 2pts

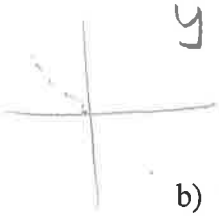
2pts 6. 5. a) Find the Cartesian coordinates for the polar coordinates)

$(-3, -5\pi/4)$ $(-3\sqrt{2}/2, -3\sqrt{2}/2)$ $(2, -\pi/6)$ $(\sqrt{3}, -1)$

$x = -3\cos(-5\pi/4)$
$-3(+\sqrt{2}/2) = -3\sqrt{2}/2$

$y = -3\sin(-5\pi/4)$
$-3(\sqrt{2}/2) = -3\sqrt{2}/2$

$x = 2\cos(-\pi/6) = 2(\sqrt{3}/2) = \sqrt{3}$
$y = 2\sin(-\pi/6) = 2(-1/2) = -1$

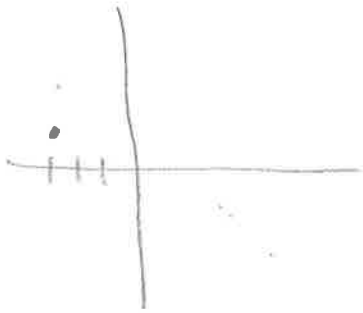


b) Find polar coordinates for the Cartesian coordinates $r^2 = x^2 + y^2$ $\tan\theta = y/x$

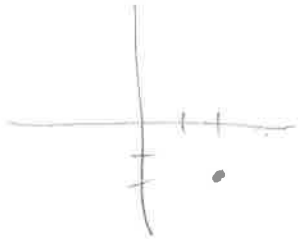
2pts $(-3, \sqrt{3})$) where $r < 0, and\ \theta > 0$ $(-2\sqrt{3}, \frac{11\pi}{6})$

2pts $(2, -2)$) where $r > 0, and\ \theta < 0$ $(2\sqrt{2}, -\pi/4)$

$r^2 = 9 + 3 = 12$
$r = 2\sqrt{3}$
$\tan\theta = \sqrt{3}/-3$
$\theta = \pi/6$

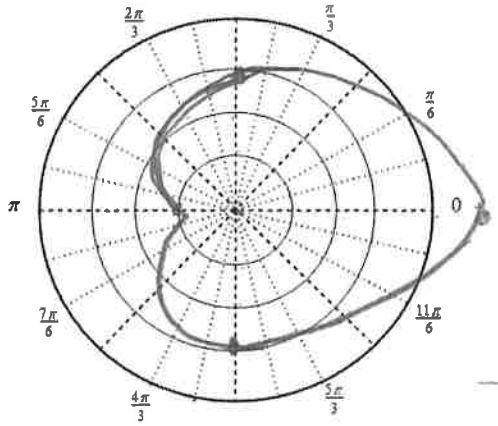


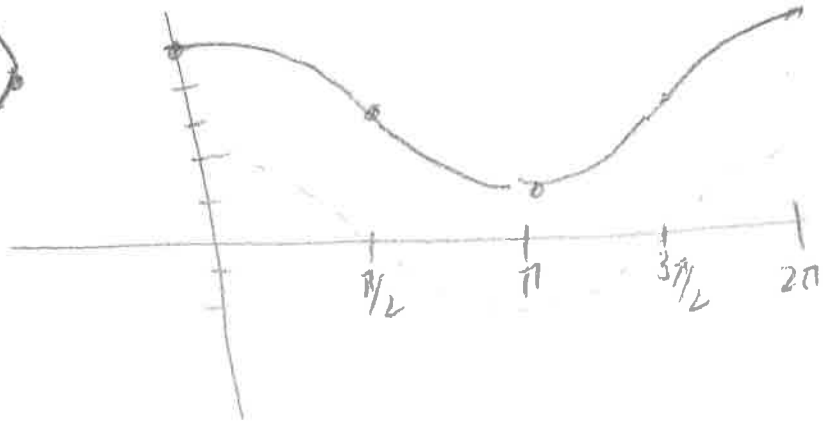
$r^2 = 4 + 4 = 8$
$r = 2\sqrt{2}$
$\tan\theta = -2/2 = -1$
$\theta = \pi/4$



7. **Sketch the graph** $r = 3 + 2\cos\theta$. .

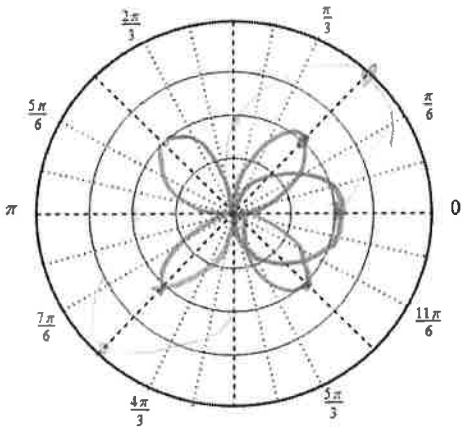
2pts





8. Find all points of intersection for the curves $r = 2\sin 2\theta$ *and* $r = 2\cos\theta$ GRAPH

2pt



$4\sin\theta\cos\theta = 2\cos\theta$

$4\sin\theta\cos\theta - 2\cos\theta = 0$

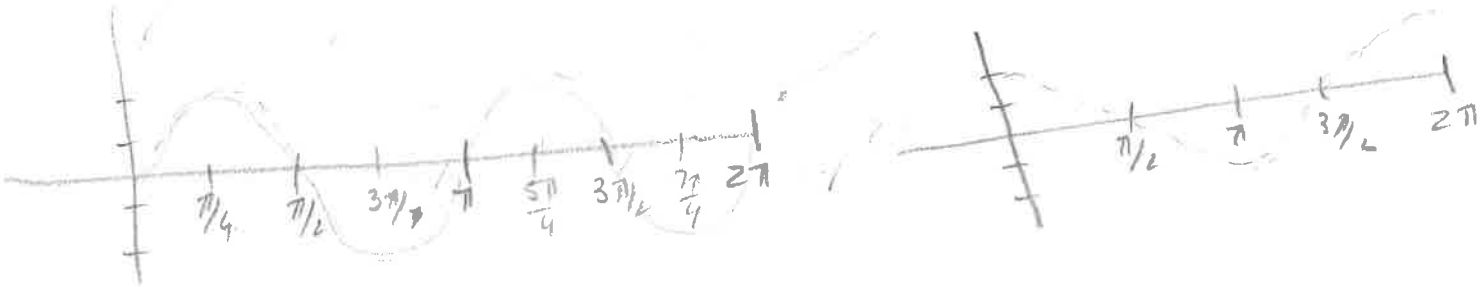
$2\cos\theta(2\sin\theta - 1) = 0$

$\cos\theta = 0$ $\quad$ $\sin\theta = 1/2$, $\theta = \frac{5\pi}{6} + 2\pi n$

$\theta = \pi/2\, n$ $\quad$ $\theta = \pi/6 + 2\pi n$

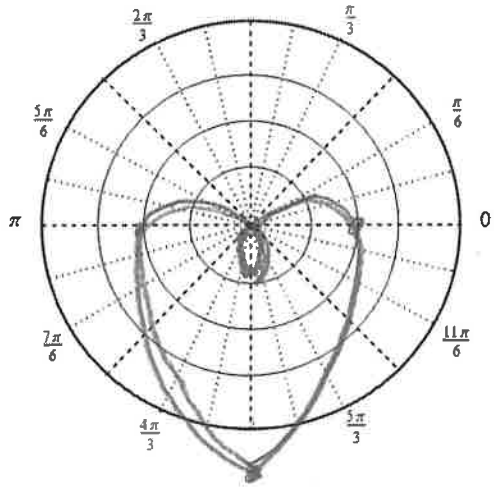
$(0, \pi/2\, n)$ $\quad$ $(\sqrt{3}, \pi/6 + 2\pi n)$

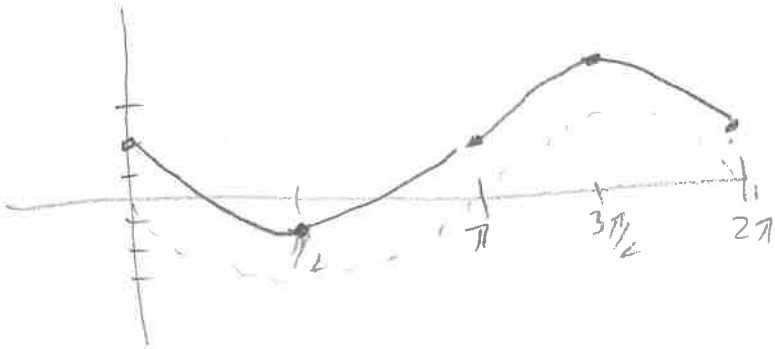
$(\sqrt{3}, \frac{5\pi}{6} + 2\pi n$



9. Sketch the graph $r = 2 - 3\sin\theta$

1pt





10. How many petals will the following curves have?

1/2pt

a) $r = 3\cos 9\theta$ 9

b) $r = 2\sin 6\theta$ 12

1/2pt

MATH 152 **QUIZ 11** NAME Solutions

Mrs. Bonny Tighe 25 points

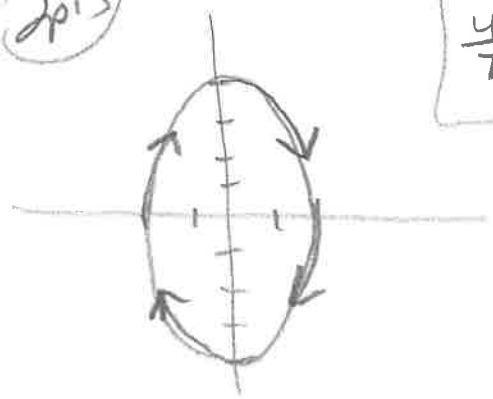
10.1, 10.2,10.3 TA_____Wed 12/4/19

1. Sketch the plane curve defined by the given parametric equations. Indicate with an arrow the direction in which the curve is traced as the parameter increases. Eliminate the parameter to find a Cartesian equation of the curve.

a) $x = 2\sin\theta,\ \ y = 4\cos\theta$

2pts

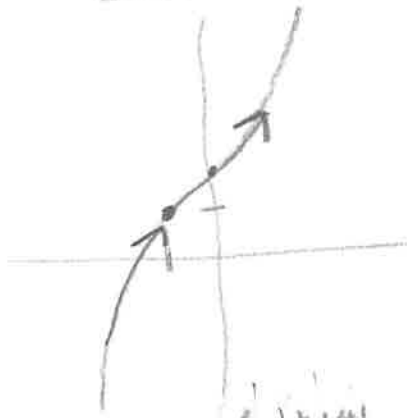
$$\boxed{\frac{y^2}{16} + \frac{x^2}{4} = 1}$$



b) $x = t - 1,\ \ y = t^3 + 1$

2pts

$t = x + 1$ $\boxed{y = (x+1)^3 + 1}$



t	x	y
0	-1	1
1	0	2

2. Find a parametric equation for the ellipse $\frac{y^2}{4} + \frac{x^2}{25} = 1$ if it travels clockwise.

1pt $x = 5\sin\theta \quad y = 2\cos\theta$

3. Find the equation of the tangent line to the curve $x = 1 + \ln t,\ \ y = t^2 + 2$ at $(1, 3)$

$x = 1,\ y = 3$ so $t = 1$

$dy/dx = \dfrac{dy/dt}{dx/dt} = \dfrac{2t}{\frac{1}{t}} = 2t^2$

$m = 2$

1pt

$$\boxed{y - 3 = 2(x - 1)}$$

$y=0$

4. Find the area enclosed by the x- axis and the curve $x = 1 + t^2$, $y = t - t^2$

$0 = t(t-1) \quad t = 1, 0$

2pts $\int_0^1 y\,dx = \int_0^1 (t - t^2)\,2t\,dt$

$\int_0^1 (2t^2 - 2t^3)\,dt = \frac{2}{3}t^3 - \frac{1}{2}t^4 \Big|_0^1 =$

$\frac{2}{3} - \frac{1}{2} = \frac{1}{6}$

5. Compute the surface area obtained by rotating the curve

3pts $x = 3t - t^3$, $y = 3t^2$, $0 \le t \le 1$ about the x-axis. $S = 2\pi \int_\alpha^\beta radius(arc\ length)$

$r = y$

$2\pi \int_0^1 3t^2 \sqrt{(3-3t^2)^2 + (6t)^2}\,dt$

$2\pi \int_0^1 3t^2 \sqrt{9 - 18t^2 + 9t^4 + 36t^2}\,dt$

$2\pi \int_0^1 3t^2 \sqrt{9 + 18t^2 + 9t^4}\,dt = 2\pi \int_0^1 3t^2 \sqrt{(3+3t^2)^2}\,dt$

$2\pi \int_0^1 3t^2 (3 + 3t^2)\,dt = 2\pi \int_0^1 (9t^2 + 9t^4)\,dt$

$2\pi \left[3t^3 + \frac{9}{5}t^5\right]_0^1 = 2\pi \left[3 + \frac{9}{5}\right]$

$2\pi \left(\frac{24}{5}\right)$

6. 5. a) Find the Cartesian coordinates for the polar coordinates)

2pts

$(-2, -5\pi/3)$ $(-1, -\sqrt{3})$ $(3, -\pi/4)$ $(3\sqrt{2}/2, -\frac{3\sqrt{2}}{2})$

2pts

$x = r\cos\theta$

$y = r\sin\theta$

$x = -2\cos(-5\pi/3)$

$-2(1/2) = -1$

$y = -2\sin(-5\pi/3) = -2(\sqrt{3}/2)$

$x = 3\cos(-\pi/4) = 3\sqrt{2}/2$

$y = 3\sin(-\pi/4) = -3\sqrt{2}/2$

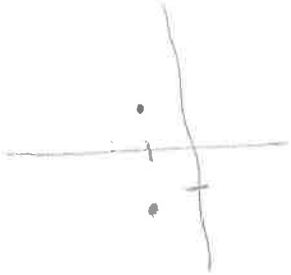
b) Find polar coordinates for the Cartesian coordinates

2pts

$(-1, -\sqrt{3})$) where $r < 0, and\ \theta > 0$ $(-2, \pi/3)$

2pts

$(-1, 1)$) where $r > 0, and\ \theta < 0$ $(\sqrt{2}, -5\pi/4)$



$r^2 = x^2 + y^2$

$r^2 = 1 + 3 = 4$

$r = 2$

$\tan\theta = -\sqrt{3}/-1 = \sqrt{3}$

$\theta = \pi/3$

$(2, 4\pi/3)$

$\tan\theta = y/x$

$r^2 = 1 + 1 = 2$

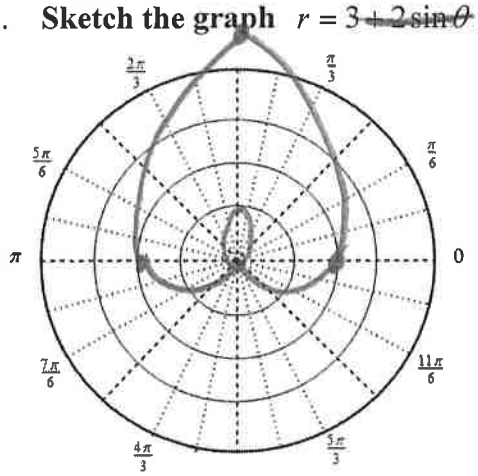
$r = \sqrt{2}$

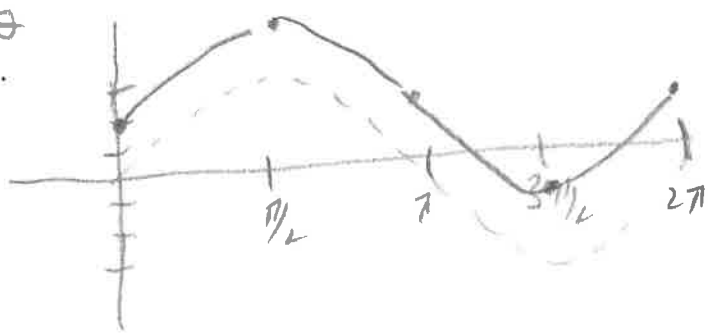
$\tan\theta = 1/-1 = -1$

$(\sqrt{2}, 3\pi/4)$

7. **Sketch the graph** $r = \cancel{3 + 2\sin\theta}$ $2 + 3\sin\theta$. .

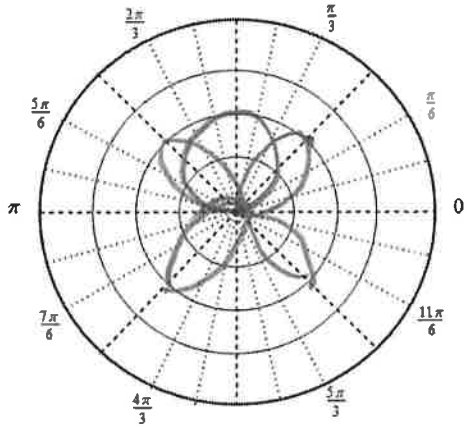
2pts





8. Find all points of intersection for the curves $r = \sin 2\theta$ *and* $r = \sin\theta$ GRAPH

2pts



$\sin 2\theta = \sin\theta$

$2\sin\theta\cos\theta - \sin\theta = 0$

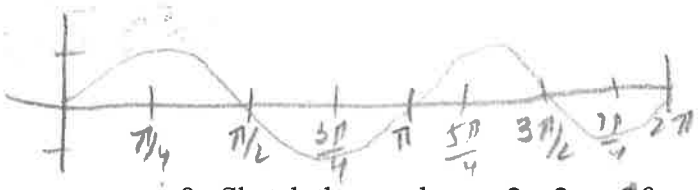
$\sin\theta(2\cos\theta - 1) = 0$

$\sin\theta = 0$ $\quad$ $\cos\theta = 1/2$

$\theta = \pi n$ $\quad$ $\theta = \pi/3 + 2\pi n$

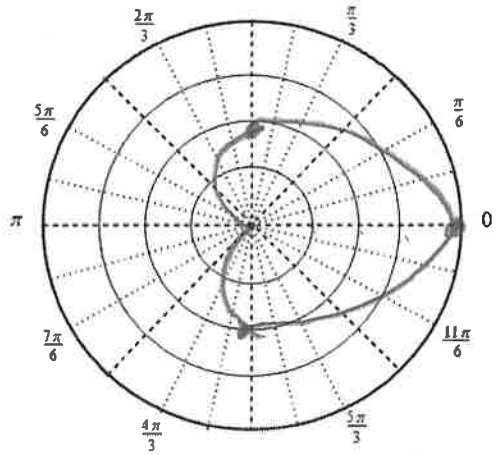
$(0, \pi n)$ $\quad$ $(\sqrt{3}/2, \pi/3 + 2\pi n)$

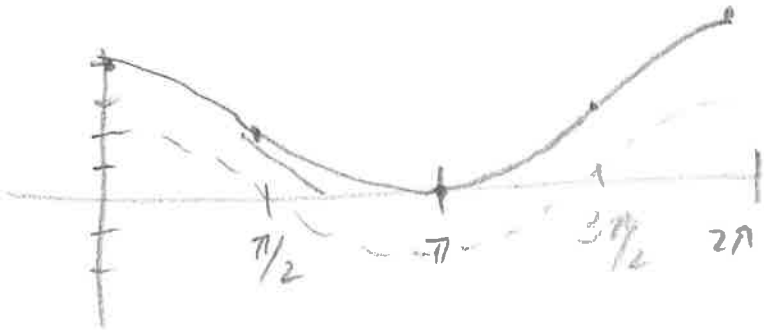
$(-\sqrt{3}/2, \frac{5\pi}{3} + 2\pi n$



9. Sketch the graph. $r = 2 + 2\cos 2\theta$.

1pt





10. How many petals will the following curves have?

$\frac{1}{2}$pt

a) $r = 3\cos 8\theta$ 16 $\qquad$ b) $r = 2\sin 11\theta$ 11 $\frac{1}{2}$pt